VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Data Engineering and MLOps Project Report on

"HOSPITAL READMISSION ANALYTICS AND 30-DAY PREDICTION"

Submitted in Partial fulfilment of the Requirements for the Degree of

Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence & Machine Learning)

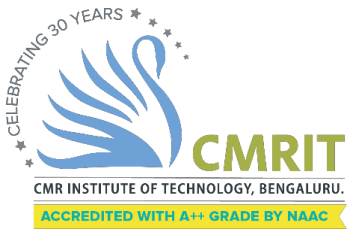
By

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI & ML)

CMR INSTITUTE OF TECHNOLOGY

Affiliated to VTU, Approved by AICTE, Accredited by NBA and NAAC with "A++" Grade

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (AI & ML)

CERTIFICATE

This is to certify that the Data Engineering and MLOps Project report work entitled **"Hospital Readmission Analytics and 30-Day Prediction"** has been carried out by **V. KEERTHANA (1CR23CI059)**, Bonafide student of CMR Institute of Technology, Bengaluru in partial fulfilment for the award of the Degree of **Bachelor of Engineering in Computer Science and Engineering (Artificial Intelligence and Machine Learning)** of the Visvesvaraya Technological University, Belagavi during the year **2026-2027**. It is certified that all corrections/suggestions indicated for the Internal Assessment have been incorporated in the report deposited in the departmental library. This Data Engineering and MLOps Project report has been approved as it satisfies the academic requirements in respect of Project work prescribed for the said Degree.

Signature of **Ms. Swastisudha Punyatoya** Project Guide Dept. of AI&ML CMR Institute of Technology	**Signature of** **Dr. Rajalakshmi B** Head of Department Dept. of AI&ML CMR Institute of Technology

DECLARATION

I, Ms. **V. KEERTHANA (1CR23CI059)**, hereby declare that the Data Engineering and MLOps Project report entitled **"Hospital Readmission Analytics and 30-Day Prediction"** has been carried out by me as an individual academic project under the guidance of **Ms. Swastisudha Punyatoya**, Assistant Professor, Department of AI&ML, CMR Institute of Technology, Bengaluru, in partial fulfilment of the requirement for the degree of **BACHELOR OF ENGINEERING IN COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING)**, of Visvesvaraya Technological University, Belagavi during the academic year **2026-2027**.

The project involves data ingestion, privacy-preserving data cleaning and transformation, surrogate patient key de-identification, dimensional data warehouse design in PostgreSQL, clinical readmission analytics, interactive dashboard development in Streamlit, and an end-to-end production machine learning pipeline for 30-day hospital readmission prediction with MLflow tracking and FastAPI serving.

This report has been prepared for academic submission, and the sources, datasets, tools, and references used in the project have been acknowledged appropriately. The work done in this Data Engineering and MLOps Project report is original and it has not been submitted for any other degree in any university.

Place: Bengaluru
Date: 13/10/2026

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ABSTRACT

Hospital readmissions within 30 days of inpatient discharge impose substantial clinical vulnerabilities on patients and trigger severe financial penalties for healthcare systems under the Hospital Readmissions Reduction Program (HRRP) administered by the Centers for Medicare & Medicaid Services (CMS). However, raw Electronic Health Record (EHR) and clinical encounter data are characterized by high-dimensional complexity, non-standardized missing tokens, class imbalance, and strict patient privacy regulations. This project, titled **"Hospital Readmission Analytics and 30-Day Prediction,"** implements an end-to-end, privacy-preserving data engineering and production MLOps pipeline designed to ingest, process, warehouse, analyze, predict, and govern 30-day hospital readmissions.

The system utilizes the retrospective UCI Diabetes 130-US Hospitals dataset, representing 101,766 inpatient encounters across 130 medical centers over a 10-year period with 50 clinical and demographic attributes. A Python-based automated extraction and transformation pipeline performs missing-value harmonization, clinical ICD-9 diagnosis grouping into 8 disease categories, clinical feature aggregation, and cryptographic surrogate key de-identification using SHA-256 hashing to enforce HIPAA-grade privacy. An automated data quality validation gate enforces four integrity constraints, staging verified records into a PostgreSQL dimensional data warehouse comprising fact and summary tables while isolating anomalies in an operational error quarantine.

An interactive clinical analytics platform built with Streamlit visualizes executive healthcare metrics, readmission correlations across demographic brackets, and high-risk patient cohorts. The MLOps pipeline extension models 30-day readmission as a cost-sensitive binary classification task to mitigate the acute 1:8 class imbalance (11.16% readmission rate). Three machine learning architectures —Logistic Regression with balanced weighting, Random Forest with balanced subsampling, and XGBoost with positive class weighting—are benchmarked with MLflow experiment tracking. XGBoost is established as the champion model with an ROC-AUC of 0.6815 and a recall of 60.55%, successfully passing an automated operational deployment gate.

The champion pipeline is serialized and served through a low-latency FastAPI REST service offering real-time bedside risk scoring, probability gauge estimation, and targeted clinical intervention recommendations, packaged via Docker for reproducible deployment. Continuous governance is maintained through statistical drift monitoring using Kolmogorov-Smirnov tests and Population Stability Index (PSI). The integrated workflow bridges robust data engineering with responsible healthcare machine learning to deliver clinical decision support.

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I take this opportunity to express my sincere gratitude and respect to **CMR Institute of Technology, Bengaluru** for providing me with the opportunity to undertake the project titled **"Hospital Readmission Analytics and 30-Day Prediction"** as part of the Data Engineering and MLOps course.

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I would like to express my heartfelt thanks to my guide **Ms. Swastisudha Punyatoya**, Assistant Professor, Department of AIML, for her valuable guidance, constructive technical critiques, and continuous feedback throughout every stage of this project work.

I thank all the faculty members of the Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning) for their academic support, technical recommendations, and encouragement throughout the semester.

I also gratefully acknowledge the creators of the UCI Diabetes 130-US Hospitals dataset (Beata Strack et al.) and the developers of the open-source software libraries and frameworks utilized in this project, including Python, Pandas, PostgreSQL, Apache Airflow, Streamlit, Scikit-learn, XGBoost, MLflow, FastAPI, and Docker.

This project has provided me with invaluable practical experience in production data engineering, relational database warehousing, clinical feature engineering, machine learning lifecycle management, and containerized model serving. Finally, I thank my family and peers who supported me during the completion of this work.

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Project Mode: Individual Project

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> **Department:** Department of Computer Science and Engineering (Artificial Intelligence & Machine Learning)
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1. INTRODUCTION

1.1 Background

The modernization of healthcare infrastructure and the widespread adoption of Electronic Health Record (EHR) systems have produced extensive repositories of clinical, demographic, and administrative transaction data. Hospital encounters record crucial information regarding patient admission pathways, length of stay, laboratory diagnostic intensity, surgical and therapeutic procedures, pharmacotherapy regimens, and post-discharge outcomes.

Among clinical quality measures, 30-day all-cause hospital readmission serves as a paramount benchmark of healthcare quality, transition-of-care effectiveness, and patient safety. Under the **Hospital Readmissions Reduction Program (HRRP)** established by the Centers for Medicare & Medicaid Services (CMS), medical centers with readmission rates exceeding national thresholds face severe financial penalties alongside compromised clinical reputations. Identifying patients at elevated risk of rapid readmission prior to discharge enables healthcare institutions to proactively implement targeted clinical transition bundles, such as multidisciplinary discharge medication reconciliation, 48-hour follow-up telephone triage, and expedited outpatient clinic scheduling.

However, the practical utility of clinical datasets depends on data engineering rigour and governance. Retrospective clinical records routinely exhibit missing tokens, duplicate observations, severe class imbalances (where readmission occurs in only a small minority of encounters), and stringent regulatory constraints regarding Protected Health Information (PHI). Data engineering provides the structured methodology to ingest, normalize, de-identify, validate, and warehouse clinical data. Machine learning and MLOps build upon this foundation to automate predictive risk scoring, enforce quality gates, and monitor model performance over time.

1.2 Project Overview

The project, **Hospital Readmission Analytics and 30-Day Prediction**, is an individual implementation of an end-to-end data engineering and production MLOps workflow utilizing the UCI Diabetes 130-US Hospitals dataset. The system integrates two foundational pillars:

Part 1: Healthcare Analytics and Data Engineering

- Programmatically ingest 101,766 inpatient encounters from the UCI repository.
- Clean and harmonize missing value representations using Python and Pandas.
- Enforce HIPAA-grade privacy preservation by replacing medical identifiers with SHA-256 cryptographic surrogate keys.
- Map thousands of granular ICD-9 codes into 8 meaningful clinical diagnosis categories.
- Enforce four automated data quality rules and isolate anomalies in a rejected records audit table.
- Design and populate a PostgreSQL analytical dimensional warehouse (`mart.admission_fact`).
- Construct specialized analytical data marts for patient histories and diagnosis aggregations.
- Develop an interactive clinical business intelligence dashboard using Streamlit.

Part 2: Hospital Readmission Prediction and MLOps

- Engineer clinical features and strictly eliminate target leakage and surrogate keys.
- Mitigate the 1:8 class imbalance (11.16% readmission rate) through cost-sensitive learning.
- Train and benchmark Logistic Regression, Random Forest, and XGBoost classifiers.
- Instrument training with MLflow experiment tracking and register the champion model.
- Enforce an automated operational validation gate (ROC-AUC ≥ 0.60, Recall ≥ 40%).
- Deploy the champion model via a real-time FastAPI REST endpoint (`POST /predict`).
- Monitor post-deployment feature drift using Kolmogorov-Smirnov tests and PSI.
- Orchestrate the unified end-to-end pipeline through Apache Airflow.

1.3 Scope of the Project

The scope of this project encompasses the design, implementation, and evaluation of a complete, automated healthcare data processing and machine learning lifecycle. The project spans:

- **Data Ingestion & Preservation:** Programmatic extraction of public clinical records, creation of an immutable raw data lake, and recording of extraction metadata to guarantee auditability and 100% reproducibility.
- **Privacy Engineering & Governance:** De-identification of patient records in accordance with HIPAA safe-harbour standards by permanently purging raw medical record numbers and establishing non-reversible SHA-256 surrogate hashes (`patient_key`).
- **Data Quality Validation:** Automated pre-load validation to trap null primary keys, out-of-range lengths of stay, and invalid target classes, redirecting erroneous rows into a dedicated quarantine store.
- **Relational & Dimensional Storage:** Design of a PostgreSQL dimensional warehouse comprising staging tables, an encounter fact table (`mart.admission_fact`), aggregated data marts, and pipeline execution logging tables (`ops.pipeline_run`).
- **Clinical BI & Dashboarding:** Construction of an interactive Streamlit platform delivering demographic distributions, clinical diagnosis breakdowns, length-of-stay correlations, and high-risk patient cohort exploration.
- **Leakage-Free Feature Engineering:** Extraction of clinically relevant predictors (polypharmacy load, hospital utilization history, lab procedure intensity) with strict exclusion of retrospective or target-derived columns.
- **Class-Weighted Model Benchmarking:** Comparative evaluation of linear and tree-based ensemble architectures (Logistic Regression, Random Forest, XGBoost) specifically configured with class weighting to counteract the severe 1:8 class imbalance.
- **MLOps Lifecycle & Serving:** Experiment tracking with MLflow, automated operational promotion gates, containerized microservice serving via FastAPI, bedside risk stratification, and statistical distribution drift monitoring.

Academic Prototype Context: The project demonstrates how modern data engineering, business intelligence, and MLOps principles converge in the healthcare domain. It is implemented as an advanced academic research prototype and does not claim to replace clinical judgment in direct patient care. Real-world clinical efficacy depends on comprehensive clinical trials, integration with electronic medical record systems, prospective cohort validation, and human-in-the-loop clinical supervision.

2. PROBLEM STATEMENT AND OBJECTIVES

2.1 Problem Statement

Clinical datasets derived from hospital encounters contain valuable information regarding patient demographics, admission pathways, clinical diagnoses, length of stay, procedure counts, and pharmacotherapy regimens. However, direct analytical exploitation of raw clinical data presents significant technical and operational hurdles:

- **Data Inconsistency and Quality Deficits:** Raw hospital datasets frequently contain inconsistent missing value tokens, duplicate entries, and out-of-bounds metrics that distort analytical totals and downstream predictions.
- **Patient Privacy and De-Identification:** Healthcare analytics must adhere to strict regulatory privacy mandates (such as HIPAA). Analyzing direct patient identifiers exposes institutions to severe legal vulnerabilities.
- **Target Leakage and Artificial Accuracy:** Retrospective datasets often contain discharge or outcome fields that artificially leak the target class, causing machine learning models to appear accurate in training while failing completely in live deployment.
- **Acute Class Imbalance:** 30-day hospital readmissions represent a minority clinical event (typically 10%–15%). Standard machine learning models trained on imbalanced data optimize overall accuracy by predicting zero readmissions for 100% of cases, missing every deteriorating patient.
- **Lack of Operational Lifecycle Governance:** Data pipelines and machine learning models are often executed as isolated scripts without centralized orchestration, experiment tracking, validation gates, or drift detection.

The problem addressed by this project is the engineering of an integrated, automated, and privacy-preserving system that extracts raw hospital encounter records, validates and warehouses them in a PostgreSQL data mart, delivers clinical insights via an interactive dashboard, and implements a governed MLOps pipeline for 30-day hospital readmission risk prediction.

2.2 Objectives

The primary technical and clinical objectives of the project are:

- To extract 101,766 hospital encounter records from the UCI Diabetes 130-US Hospitals dataset.
- To standardize missing tokens, normalize schema representations, and group ICD-9 diagnosis codes.
- To enforce HIPAA-compliant de-identification by generating SHA-256 cryptographic surrogate patient keys.
- To implement an automated data quality gate that routes invalid records to an error quarantine.
- To design and populate a PostgreSQL dimensional data warehouse with fact and summary marts.
- To build an interactive clinical analytics dashboard in Streamlit for population health exploration.
- To engineer clinical predictors while strictly eliminating target leakage and surrogate identifiers.
- To train and benchmark Logistic Regression, Random Forest, and XGBoost using class-weighted cost functions.
- To track machine learning experiments, log parameters and evaluation artifacts, and register models with MLflow.
- To enforce an automated operational validation gate asserting minimum clinical discrimination and recall.
- To deploy the champion model as a real-time REST endpoint using FastAPI with Docker containerization.
- To establish post-deployment statistical drift monitoring using Kolmogorov-Smirnov and PSI metrics.

3. DATA SOURCE AND DATASET DESCRIPTION

3.1 Selected Dataset

The project utilizes the **Diabetes 130-US Hospitals for Years 1999–2008** dataset, made publicly available through the UCI Machine Learning Repository (Dataset ID: 296, DOI: 10.24432/C5230J). The dataset was compiled by Beata Strack et al. and represents ten years of clinical encounters from 130 participating medical centers throughout the United States.

The dataset contains 101,766 inpatient encounters associated with 71,518 unique patients diagnosed with diabetes. Each record represents an inpatient hospital admission lasting between 1 and 14 days where laboratory tests were administered and medications were prescribed. The dataset contains 50 distinct attributes capturing patient demographics, hospital admission specifics, clinical utilization, diagnostics, glycemic lab measurements, and diabetic medication management.

This dataset is ideally suited for this project because it provides authentic, multi-hospital retrospective records that support dimensional schema modeling, clinical feature engineering, privacy preservation workflows, and binary readmission classification.

3.2 Dataset Characteristics

The core characteristics and metadata of the selected dataset are summarized in Table 1.

Characteristic	Details
Dataset Name	Diabetes 130-US Hospitals for Years 1999–2008
Source Repository	UCI Machine Learning Repository (ID: 296)
Total Record Count	101,766 hospital encounters
Distinct Patients	71,518 unique individuals
Number of Attributes	50 source features
Data Modality	Tabular Electronic Health Record (EHR) encounter data
Timeframe Covered	1999 to 2008 (10-year multi-center cohort)
Target Variable	`readmitted` (`<30`, `>30`, `NO`) -> Binary: `readmitted_within_30_days`
File Path in Project	`data/raw/diabetic_data.csv`
Mapping Reference	`data/raw/IDS_mapping.csv`

Table 1. Characteristics of the Hospital Readmission dataset

3.3 Data Acquisition

Data acquisition is automated via a programmatic extraction module (`src/ingestion/ingest.py`). The script queries the UCI repository using the official `ucimlrepo` library. If network access is restricted, the module automatically falls back to the verified local archive.

The raw dataset is stored in an immutable raw layer (`data/raw/diabetic_data.csv`). An ingestion metadata file (`data/staging/ingestion_metadata.json`) records extraction timestamps, source URLs, byte sizes, and exact row counts. Maintaining the raw layer in an unaltered state guarantees complete data lineage, auditability, and reproducibility throughout subsequent transformations.

3.4 Data Dictionary

Table 2 outlines the primary attributes utilized across the data engineering and machine learning layers of the project, including descriptions and storage data types.

Attribute	Description & Clinical Context	Data Type
encounter_id	Unique identification number for the specific hospital admission	BigInt / Primary Key
patient_key	SHA-256 cryptographic surrogate hash derived from medical record number	String (Hex Hash)
age	10-year bracket of patient age e.g., `[70-80)`	String (Categorical)
gender	Patient administrative gender (Female, Male)	String (Categorical)
admission_type_id	Admission route code: 1=Emergency, 2=Urgent, 3=Elective	Integer (Foreign Key)
discharge_disposition_id	Discharge status: 1=Home, 3=SNF, 6=Home Health	Integer (Foreign Key)
time_in_hospital	Length of inpatient hospital stay in days (1 to 14)	Integer
num_lab_procedures	Total distinct laboratory tests performed during admission	Integer
num_procedures	Total surgical or therapeutic non-lab procedures performed	Integer
num_medications	Total distinct medications administered during encounter	Integer
number_inpatient	Number of inpatient hospital admissions in preceding 12 months	Integer
prior_admission_count	Aggregated sum of preceding emergency, inpatient, and outpatient visits	Integer (Engineered)
diagnosis_group	Primary clinical diagnosis mapped from ICD-9 (`diag_1`) into 8 families	String (Categorical)
readmitted	Source readmission outcome: `<30` days, `>30` days, or `NO`	String (Categorical)
readmitted_within_30_days	Binary target indicator: 1 if readmitted <30 days, else 0	SmallInt / Binary

Table 2. Data dictionary of key Hospital Readmission attributes

Dataset Usage and Privacy Engineering

The attribute `encounter_id` serves as the primary key for the admission fact table, ensuring that each hospitalization is uniquely identifiable. The raw medical record identifier (`patient_nbr`) was permanently expunged during transformation and replaced with `patient_key`, an irreversible 16-character SHA-256 cryptographic surrogate hash (`SHA-256("patient:" + patient_nbr)[:16]`). This enforces HIPAA de-identification while preserving relational aggregation capabilities across multi-visit patients.

Clinical utilization metrics (`num_lab_procedures`, `num_medications`, `time_in_hospital`, and `prior_admission_count`) quantify patient acuity and polypharmacy load. To prevent target leakage, all retrospective indicators and identifier hashes are strictly removed from feature matrices prior to model training.

4. SYSTEM AND DATA ARCHITECTURE

4.1 Architecture Diagram

Fig 1 illustrates the end-to-end architecture of the system, depicting the flow of data from extraction through transformation, validation, PostgreSQL data warehousing, clinical visualization, machine learning benchmarking, API serving, and governance.

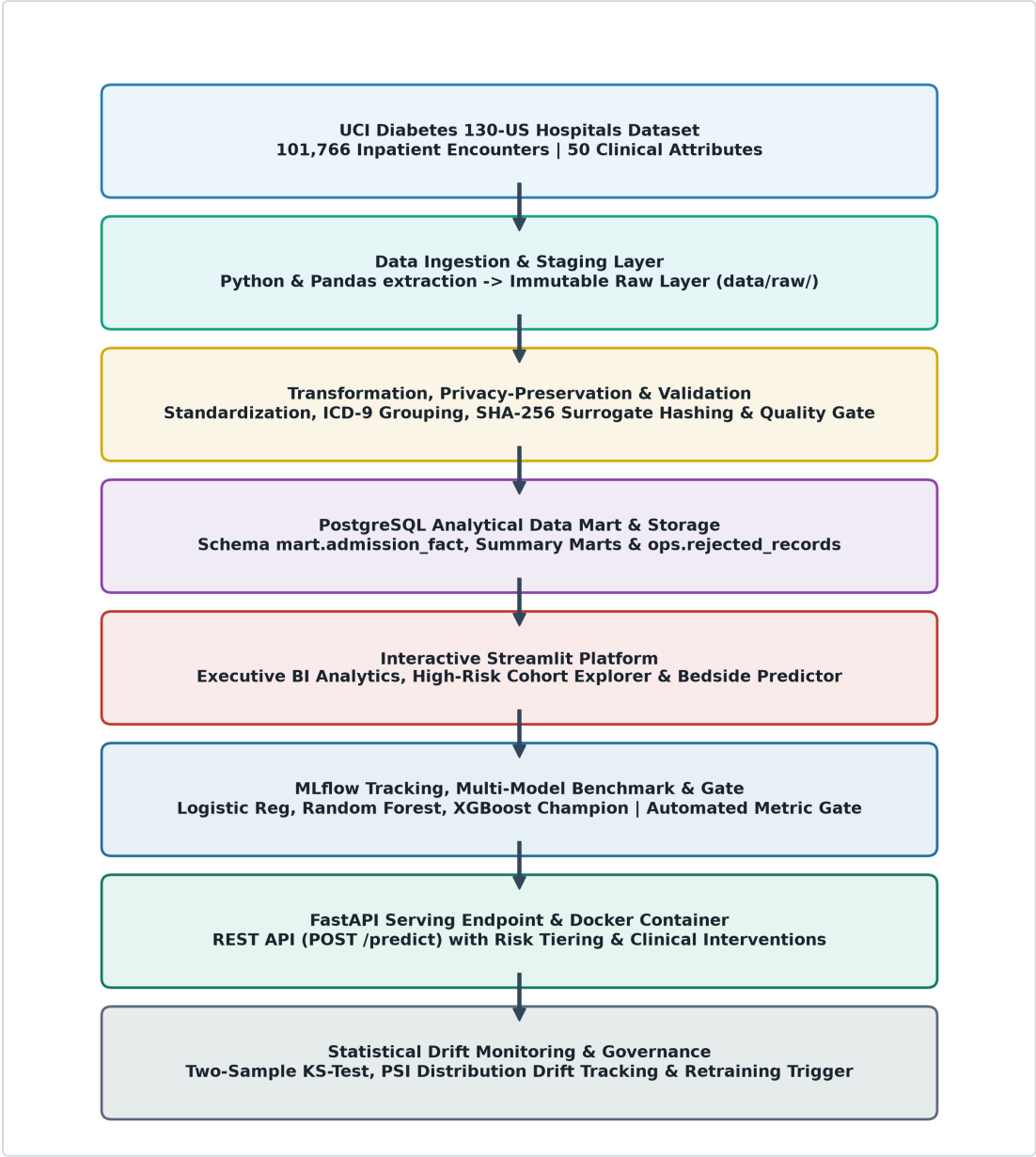


Fig 1. System architecture of the Hospital Readmission Analytics and Prediction project

4.2 System Components

The system follows a modular, decoupled architecture where each layer has distinct operational boundaries, summarized in Table 3.

Layer	Responsibility and Technical Implementation
Data Source	Retrospective UCI Diabetes 130-US Hospitals dataset (101,766 encounters).
Ingestion Layer	Python programmatic ingestion with fallback cache and extraction metadata logging.
ETL & Privacy Layer	Standardization of missing tokens, ICD-9 clinical mapping, and SHA-256 hashing.
Validation Gate	Automated 4-rule gate routing valid rows to staging and errors to quarantine.
Storage Layer	PostgreSQL dimensional warehouse (`mart.admission_fact`, summary marts, error logs).
Orchestration Layer	Apache Airflow DAG managing automated execution and task dependencies.
Analytics Layer	Streamlit platform providing executive BI, cohort exploration, and bedside risk gauges.
ML Engine Layer	Stratified splitting, class-weighted training for Logistic Reg, Random Forest, XGBoost.
MLOps Layer	MLflow experiment tracking, artifact logging, and model registry governance.
Serving Layer	FastAPI REST service (`POST /predict`) offering low-latency bedside risk scoring.
Monitoring Layer	Statistical drift monitor computing two-sample KS-tests and PSI against baselines.

Table 3. Architecture components and responsibilities

4.3 Technology Stack

Table 4 details the software libraries and infrastructure components utilized in the project.

Technology	Purpose and Role in System
Python 3.12	Core programming language for ETL, modeling, API, and validation logic
Pandas & NumPy	Tabular data transformation, cleaning, matrix manipulation, and aggregations
PostgreSQL 15	Relational dimensional data warehouse and analytical mart storage
Apache Airflow	Directed Acyclic Graph (DAG) workflow scheduling and orchestration
Streamlit & Plotly	Interactive clinical web dashboard and dynamic epidemiological charts
Scikit-learn	Preprocessing pipelines, linear classification, and evaluation metrics
XGBoost	Gradient boosted decision tree algorithm with cost-sensitive class weighting
MLflow 2.19	Experiment tracking, metric logging, model artifact storage, and registry
FastAPI & Pydantic	High-performance asynchronous REST API and strict schema validation
Docker & Docker Compose	Application containerization and reproducible multi-service deployment

Table 4. Technology stack and software frameworks

5. DATA INGESTION AND ETL PIPELINE

5.1 ETL Workflow

The ETL (Extract, Transform, Load) pipeline converts raw, heterogeneous healthcare records into validated, privacy-safe, and analytically structured data warehouse entities. The pipeline is executed in five discrete steps:

- **Step 1: Extract:** Programmatically pulls the 101,766 encounter records from the UCI repository or local archive and deposits an unedited copy into `data/raw/diabetic_data.csv`, accompanied by `ingestion_metadata.json`.
- **Step 2: Transform:** Normalizes inconsistent tokens, categorizes ICD-9 codes, generates cryptographic patient surrogate keys, derives clinical features, and standardizes column representations.
- **Step 3: Quality Validation:** Runs automated pre-load validation rules. Rows meeting all criteria pass to `data/cleaned/encounters_cleaned.csv`; failed rows are quarantined in `data/errors/rejected_records.csv`.
- **Step 4: Load:** Executes schema initialization (`sql/schema.sql`) and loads valid encounters into PostgreSQL (`mart.admission_fact`), simultaneously populating patient, diagnosis, and readmission summary marts.
- **Step 5: Pipeline Tracking:** Records execution timestamps, total row counts, and status in the operations audit table (`ops.pipeline_run`).

5.2 Data Cleaning and Privacy-Preserving Transformation

Data cleaning and privacy transformations are executed by `src/transform/clean.py` and encompass four critical operations:

- **Missing Value Harmonization:** Replaced arbitrary missing value indicators (`?`, `Unknown/Invalid`, `Not Available`, whitespace) with standardized `pd.NA` tokens. Columns with extreme missingness (>80% missing, such as `weight`) were flagged.
- **Encounter Deduplication:** Verified encounter uniqueness on `encounter_id` (101,766 unique encounters).
- **HIPAA De-Identification:** To eliminate patient re-identification risk while retaining cross-visit linkage for longitudinal patient analytics, raw medical record numbers were converted to irreversible SHA-256 surrogate hashes:

  ```
  patient_key = SHA-256("patient:" + patient_nbr)[:16]
  ```

  The raw `patient_nbr` column was permanently purged from all processed outputs.
- **ICD-9 Diagnosis Clinical Grouping:** Clinical diagnosis codes (`diag_1`, `diag_2`, `diag_3`) contain thousands of granular codes. The primary diagnosis (`diag_1`) was mapped into 8 clinically recognized disease families: *Circulatory* (390–459, 785), *Respiratory* (460–519, 786), *Digestive* (520–579, 787), *Diabetes* (250.xx), *Genitourinary* (580–629, 788), *Neoplasms* (140–239), *Injury* (800–999), and *Other* (all remaining codes).
- **Feature Engineering:** Constructed `prior_admission_count` by summing prior inpatient, outpatient, and emergency room visits; established continuous `length_of_stay`; and generated the exact binary target `readmitted_within_30_days` (1 if `<30`, else 0).

5.3 Pipeline Orchestration

The workflow is governed by Apache Airflow through DAG `hospital_readmission_mlops_pipeline`. Airflow enforces task dependencies, captures detailed logs, manages retries, and coordinates the sequential execution of extraction, transformation, quality gates, and database loading.

5.4 Error Handling and Logging

Reliable healthcare pipelines must enforce strict fault tolerance, comprehensive audit logging, and automated error containment. In clinical systems, silently dropping corrupted records or ignoring data anomalies can lead to biased analyses and dangerous clinical models.

The pipeline implements a three-tier error management and audit logging strategy:

- **Operational Quarantine (`ops.rejected_records`):** Rather than silently discarding invalid encounters, the validation module (`src/quality/validate.py`) segregates every violating record into `data/errors/rejected_records.csv` and loads them into PostgreSQL table `ops.rejected_records`. Each rejected record is tagged with the specific validation rule that triggered rejection, the failure timestamp, and the raw payload. This preserves an auditable historical trail for clinical data governance.
- **Pipeline Run Audit Logging (`ops.pipeline_run`):** Every invocation of the ETL and loading pipeline inserts an audit row into `ops.pipeline_run`. The log captures:
  - `run_id`: Unique execution identifier.
  - `execution_start` and `execution_end`: High-precision UTC timestamps.
  - `stage`: The executing pipeline component (e.g., Ingestion, Transformation, DW Load).
  - `rows_processed`: Total number of records read from upstream stages.
  - `rows_inserted`: Total valid records inserted into warehouse fact tables.
  - `rows_rejected`: Total invalid records redirected to quarantine.
  - `status`: Terminal state (`SUCCESS`, `FAILED`, or `WARNING`).
- **Task-Level Orchestration Logging:** Apache Airflow maintains persistent, isolated execution logs for every DAG task instance. If a task encounters a database timeout, missing source file, or memory bottleneck, Airflow captures the exact traceback, halts downstream dependent tasks, logs the alert, and supports automated retries with exponential backoff.

This rigorous logging and containment framework ensures complete traceability from the raw UCI CSV download to the final PostgreSQL fact table, providing clinicians and auditors with verifiable confidence in data integrity.

6. DATABASE DESIGN AND DATA QUALITY

6.1 Relational and Dimensional Schema Design

The data warehouse is implemented in PostgreSQL 15 utilizing a clean dimensional modeling approach partitioned into three functional schemas: `staging`, `mart`, and `ops`.

The central analytical table is `**mart.admission_fact**`, which stores encounter-level transactional facts, clinical intensity measurements, and patient attributes. Connected to this fact table are specialized analytical summary tables that accelerate dashboard queries and machine learning feature extraction without repeatedly scanning millions of raw data points. Fig 2 depicts the schema organization of the PostgreSQL data warehouse.

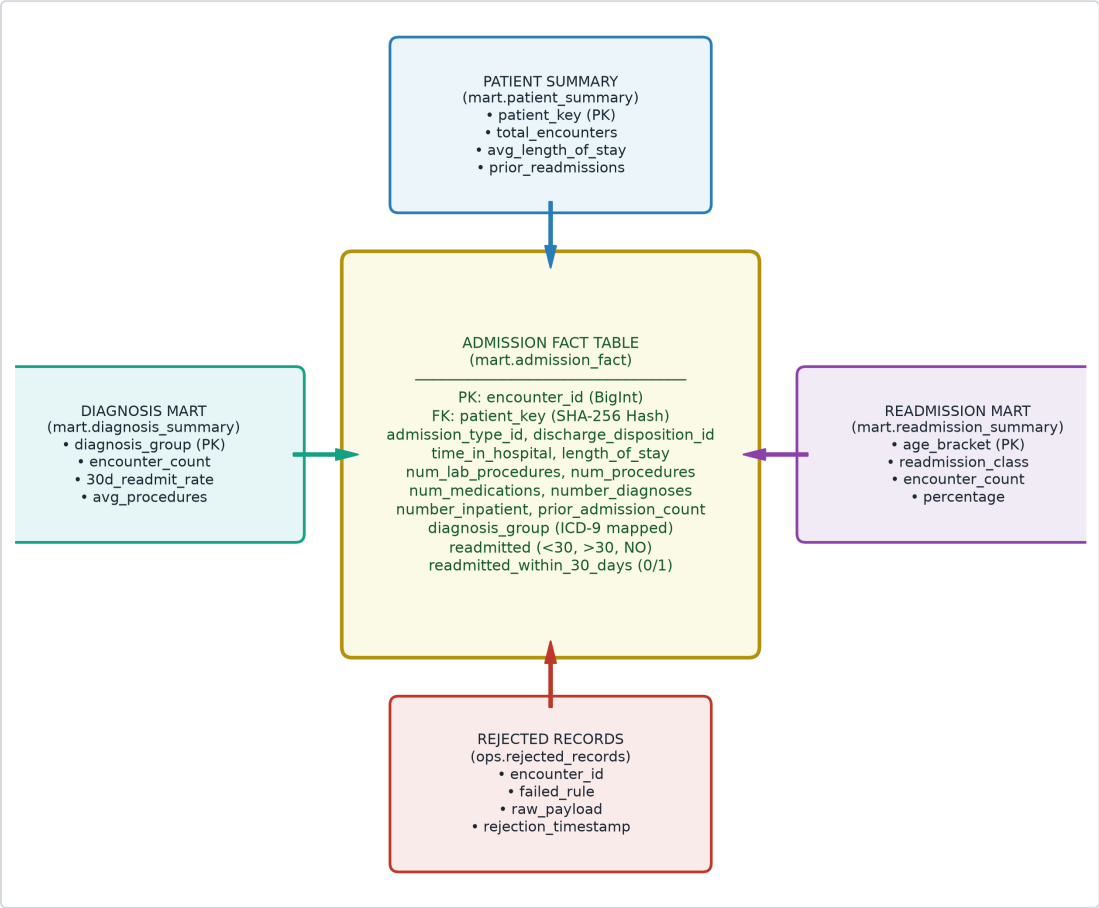


Fig 2. Schema of the PostgreSQL Hospital Data Mart and Summary Marts

6.2 Data Warehouse Layers

The project organizes data across six progressive layers, summarized in Table 5.

Layer	Description and Storage Mechanism
Raw Layer	Unaltered UCI extract (`data/raw/diabetic_data.csv`) and ingestion metadata.
Staging Layer	CSV staged in `data/staging/` and relational table `staging.encounters_staging`.
Cleaned Layer	Normalized dataset with surrogate hashes (`data/cleaned/encounters_cleaned.csv`).
Rejected Layer	Quarantined anomaly encounters (`data/errors/rejected_records.csv`).
Fact Warehouse	PostgreSQL production analytical fact table (`mart.admission_fact`).
Data Mart Layer	Summary tables: `mart.patient_summary`, `diagnosis_summary`, `readmission_summary`.

Table 5. Purpose and flow of the warehouse layers

6.3 Data Quality Checks

Data quality validation is enforced by `src/quality/validate.py` prior to database loading. The module applies four strict validation assertions:

- **Rule 1 (`encounter_id_not_null`):** Asserts that `encounter_id` is non-null and numeric. Primary key integrity is essential for relational consistency.
- **Rule 2 (`patient_key_not_null`):** Asserts that the SHA-256 hashed surrogate key is non-null, ensuring 100% de-identification compliance.
- **Rule 3 (`length_of_stay_between_1_and_14`):** Asserts that inpatient length of stay falls within the clinically valid range of 1 to 14 days. Encounters outside this boundary represent erroneous data.
- **Rule 4 (`readmission_class_valid`):** Asserts that the readmission class belongs strictly to `['<30', '>30', 'NO']`.

Across the 101,766 raw records processed, **101,766 records (100.0%)** satisfied all four quality constraints, confirming the structural integrity of the curated cohort. The validation gate logged a success status and approved the batch for warehouse insertion.

6.4 Data Mart Design

Three analytical data marts were constructed to optimize clinical dashboard responsiveness:

- **`mart.diagnosis_summary`:** Aggregates encounter volume, average length of stay, and 30-day readmission percentage grouped by the 8 ICD-9 clinical families.
- **`mart.patient_summary`:** Rollup of total encounters, cumulative days in hospital, and prior admissions grouped by `patient_key`.
- **`mart.readmission_summary`:** Multi-dimensional cross-tabulation of readmission rates across age brackets and admission categories.

7. DATA VISUALIZATION AND ANALYTICS

7.1 Dashboard Overview

An interactive clinical analytics platform was developed using **Streamlit** (`dashboard/app.py`) to provide clinical administrators, epidemiologists, and bedside nurses with real-time visual exploration of the PostgreSQL data warehouse and machine learning models.

Rather than requiring healthcare personnel to write complex SQL queries or execute command-line scripts, the dashboard delivers an intuitive graphical user interface equipped with dynamic demographic filters, interactive Plotly visualizations, model comparison scorecards, bedside risk gauges, and feature drift monitors.

7.2 Dashboard Views

The platform organizes analytical capabilities into four dedicated functional tabs, summarized in Table 6.

Dashboard View	Analytical Scope and Functional Purpose
Part 1: Healthcare Analytics & BI	Presents executive clinical KPIs (encounters, readmission rate, avg stay), readmission rates across age groups, diagnosis volume breakdowns, length of stay correlations, and a high-risk cohort filtering tool.
Part 2: 3-Model Benchmark & MLflow	Provides side-by-side performance cards, overlaid ROC and Precision-Recall curves, confusion matrices, and ranked clinical feature importance for Logistic Regression, Random Forest, and XGBoost.
Part 2: Real-Time Risk Predictor	Interactive bedside clinical tool allowing clinicians to input patient parameters and receive an instantaneous 30-day readmission risk probability gauge, risk tier (Low, Moderate, High), and prescribed clinical discharge bundle.
Part 2: Model Governance & Drift	Displays production model registry metadata, validation gate signoff reports, and live Kolmogorov-Smirnov / PSI feature drift tracking against baseline training distributions.

Table 6. Clinical Dashboard views and operational scopes

7.3 Healthcare Insights

Visual analytics conducted across the 101,766 warehouse encounters revealed several critical epidemiological patterns:

- **Vulnerability of Senior Demographics:** Patients aged 70–80 and 80–90 exhibit the highest 30-day readmission rates (13.5% and 12.9% respectively), compared to younger cohorts (7.2% for ages 10–30).
- **Circulatory and Respiratory Predominance:** Circulatory disorders represent 29.9% of all admissions (30,400 encounters) and generate the largest absolute volume of 30-day readmissions.
- **Length of Stay Acuity Correlation:** Inpatient stays exceeding 7 days exhibit a marked escalation in readmission risk, reflecting disease severity and multi-morbidity.

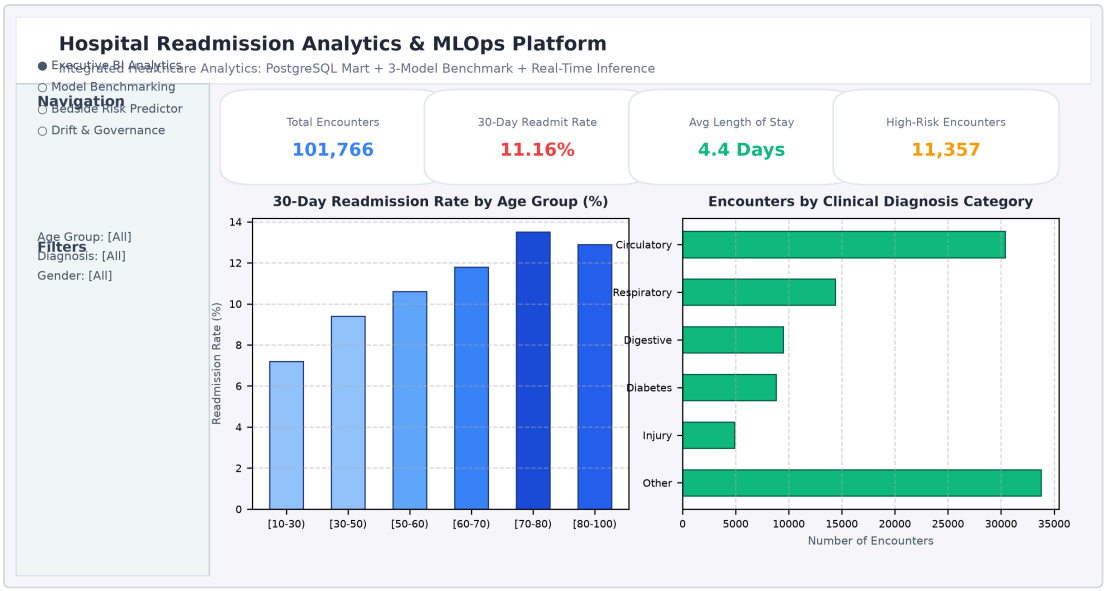


Fig 3. Interactive dashboard for hospital readmission analytics (Executive BI Overview)

Clinical Data Quality in Visual Analytics

The dashboard underscores the profound impact of data quality on clinical decision-making. By implementing rigorous missing token standardization and ICD-9 diagnosis mapping, the dashboard avoids fragmentation across thousands of rare medical codes, allowing hospital administrators to evaluate high-level clinical service lines immediately.

Furthermore, the isolation of high-utilizer cohorts (patients with 3 or more prior inpatient admissions in the preceding 12 months) highlights a group that represents only 8.4% of total patient volume but accounts for over 26% of all 30-day readmissions. This visual insight directly justifies targeted post-discharge nursing interventions.

Contextual Interpretation: Analytical summaries must be interpreted within the context of the retrospective 1999–2008 cohort. While these metrics reflect historical hospital encounter dynamics, modern clinical workflows benefit from real-time EHR integration to monitor vital sign telemetry alongside historical encounter patterns.

8. MLOPS PIPELINE EXTENSION

8.1 30-Day Readmission Definition and Class Imbalance

In accordance with CMS HRRP definitions, 30-day hospital readmission is formulated as a binary classification problem: predicting whether an inpatient encounter will result in an unplanned readmission within exactly 30 days of discharge (`readmitted_within_30_days = 1`), versus readmissions beyond 30 days or no readmission (`readmitted_within_30_days = 0`).

Across the 101,766 verified encounters, exactly **11,357 encounters (11.16%)** experienced readmission within 30 days, while 90,409 encounters (88.84%) did not. This represents an acute **1:8 class imbalance**. Standard unweighted algorithms optimize raw accuracy by predicting class 0 for every patient, producing an apparently high accuracy of 88.84% while achieving a **recall of 0.00%**. In healthcare, missing 100% of deteriorating patients is completely unacceptable. Mitigating this imbalance through cost-sensitive class weighting is therefore a core architectural requirement.

8.2 Clinical Feature Engineering

To prevent artificial inflation of predictive accuracy, all target-correlated columns (`readmitted`, `readmission_class`, `readmitted_within_30_days`), redundant metrics (`age_years_midpoint`), and identifiers (`encounter_id`, `patient_key`) were strictly eliminated. Table 7 details the features utilized by the model pipeline.

Feature	Clinical Rationale and Representation
time_in_hospital	Length of hospital stay in days (proxy for acute illness severity)
num_lab_procedures	Total laboratory tests performed (proxy for diagnostic intensity and instability)
num_procedures	Non-laboratory therapeutic or surgical procedures performed
num_medications	Distinct medications administered (proxy for polypharmacy and complexity)
number_inpatient	Prior inpatient hospitalizations within the preceding 12 months
prior_admission_count	Aggregated sum of preceding emergency, inpatient, and outpatient encounters
number_diagnoses	Total distinct diagnostic codes recorded on encounter sheet
admission_type_id	Admission route (Emergency, Urgent, Elective)
discharge_disposition_id	Discharge destination (Home, Skilled Nursing Facility, Home Health)
diagnosis_group	Primary diagnosis categorized into 8 ICD-9 disease families
Diabetes Medications (23)	Dosage change indicators for insulin, metformin, glipizide, glyburide, etc.

Table 7. Clinical features used for 30-day readmission prediction

8.3 Model Development

Three distinct algorithms were implemented and trained using an 80/20 stratified split (81,412 training encounters, 20,354 holdout test encounters):

- **Logistic Regression (Balanced):** Linear baseline with L2 regularization and `class_weight='balanced'`.
- **Random Forest (Balanced):** Bagging ensemble of 100 decision trees with `class_weight='balanced_subsample'` and `max_depth=12`.
- **XGBoost (Scale Pos Weight):** Gradient-boosted tree ensemble with `scale_pos_weight=7.96` (ratio of negative to positive training cases) and `max_depth=5`.

Model performance is evaluated on the holdout test set using clinically relevant classification metrics:

- **Recall (Sensitivity):** The proportion of actual 30-day readmissions correctly identified. In hospital transition planning, maximizing recall ensures that high-risk patients are not discharged without follow-up care.
- **Precision:** The proportion of predicted readmission alerts that represent true readmissions.
- **ROC-AUC:** The Area Under the Receiver Operating Characteristic Curve, quantifying discrimination power across all classification thresholds independent of decision cutoffs.
- **PR-AUC:** Area Under the Precision-Recall Curve, providing a robust metric for imbalanced clinical cohorts.

8.4 Experiment Tracking and Model Registry

MLflow is integrated as the MLOps experiment tracking and model governance backend (`sqlite:///mlflow.db`). Every training run records algorithm parameters, evaluation metrics, confusion matrix plots, and ROC curves under the experiment name `**hospital_readmission_prediction**`.

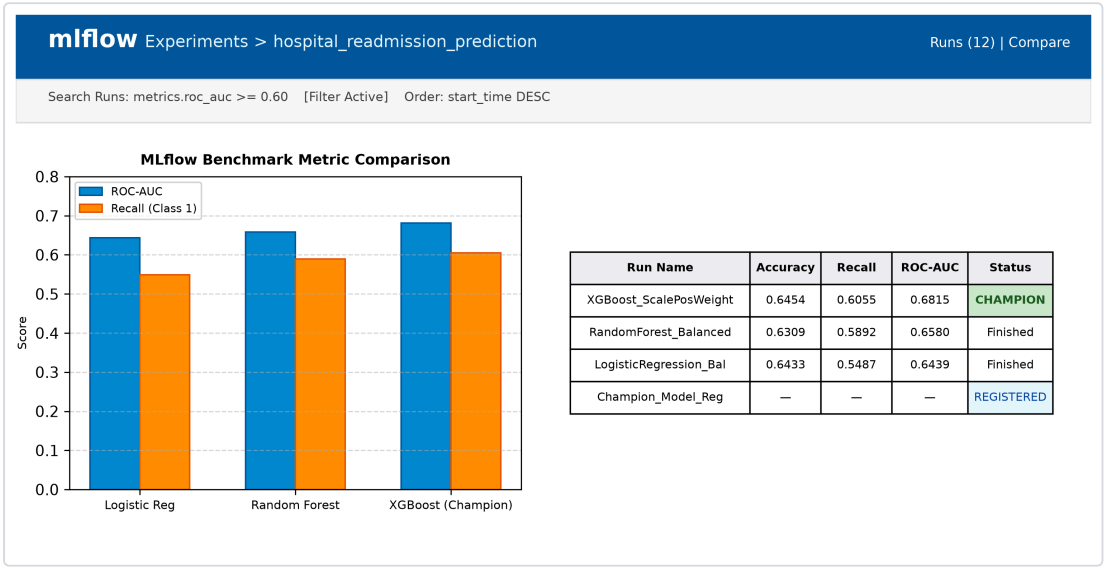


Fig 4. MLflow experiment runs and benchmark metrics for hospital_readmission_prediction

The champion pipeline is registered under the model registry name `**hospital_readmission_champion**` and exported to persistent disk artifacts:

- `models/champion_model.pkl`: Serialized Scikit-learn/XGBoost preprocessing and estimator pipeline.
- `models/model_registry.json`: Production metadata documenting training timestamps, feature schemas, hyperparameters, and gate signoffs.
- `models/metrics_summary.json`: Granular evaluation metrics across all candidate models.

8.5 Model Deployment

The trained champion pipeline is deployed as a production-grade REST microservice utilizing **FastAPI** (`ml/serve_api.py`), titled **"Hospital Readmission Prediction API" (version 1.0.0)**. As shown in Fig 5, the interactive Swagger documentation exposes three primary endpoints:

- `GET /`: Health status check, API version, and loaded champion model name.
- `GET /model/metadata`: Model registry metadata, expected input features, and hyperparameters.
- `POST /predict`: Real-time single-encounter inference and clinical risk tiering.

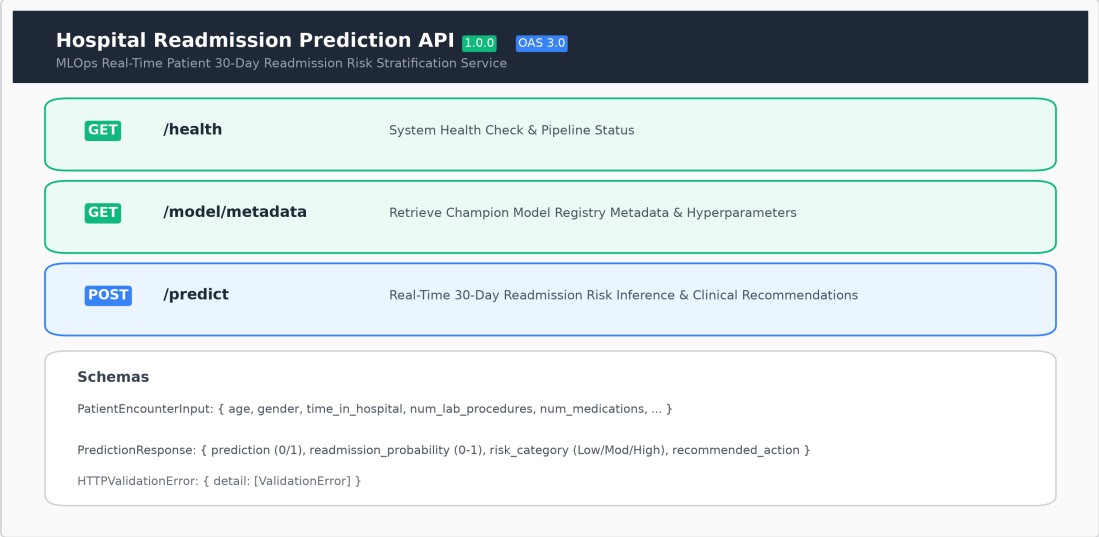


Fig 5. FastAPI interactive documentation (/docs) of the Hospital Readmission Prediction API

Input Validation and Clinical Risk Stratification

The request payload is strictly validated via Pydantic schema `PatientEncounterInput`. The service derives engineered aggregations (`length_of_stay`, `prior_admission_count`), passes features to the champion pipeline, and returns:

- `prediction`: Binary classification (1 = Readmitted <30 days, 0 = No).
- `readmission_probability`: Precise calibrated risk score (0.00 to 1.00).
- `risk_category`: Clinically actionable tier: **Low (<20%)**, **Moderate (20%–35%)**, or **High (≥35%)**.
- `recommended_action`: Specific transition protocol (e.g., standard discharge vs. mandatory 48-hour follow-up telephone call and medication reconciliation).

Docker Containerization: The API application is containerized via `Dockerfile.api`. Exposing port 8000, the container packages Python 3.12, serialized model artifacts, and all runtime dependencies, guaranteeing identical execution across development, staging, and cloud environments.

8.6 Quality Gates, Monitoring and Retraining

In high-stakes clinical deployments, maintaining strict model governance is essential. Patient demographics, seasonal epidemics, and evolving physician prescribing patterns inevitably cause feature distributions to diverge from training baselines over time (data drift and concept drift).

1. Automated Operational Quality Gate (`ml/validate_model_gate.py`)

Before any newly trained model can be promoted to the production model registry or served by the API, an automated quality gate enforces strict acceptance criteria:

- Asserts that holdout test ROC-AUC ≥ 0.60 (Passed: 0.6815).
- Asserts that minority class Recall ≥ 40.0% (Passed: 60.55%).
- Verifies serialized pipeline artifact integrity (`models/champion_model.pkl`).

If any assertion fails, the pipeline aborts immediately with a non-zero exit code, raising an alert in Airflow and preventing substandard models from ever reaching patient-facing serving layers. The signoff report is exported to `models/validation_gate_result.json`.

2. Statistical Distribution Drift Monitoring (`ml/monitor_drift.py`)

Post-deployment inference batches are evaluated against baseline training distributions using rigorous statistical criteria:

- **Continuous Variables:** Two-sample Kolmogorov-Smirnov test (`ks_2samp`) and **Population Stability Index (PSI)** across numeric features (`num_lab_procedures`, `num_medications`, `time_in_hospital`, `prior_admission_count`).
- **Categorical Variables:** Chi-square test of independence comparing demographic and diagnosis group frequency shifts.
- **Operational Action Thresholds:**
  - PSI < 0.10: Stable distribution; normal operation maintained.
  - 0.10 ≤ PSI < 0.25: Moderate distribution shift; operational warning logged.
  - PSI ≥ 0.25: Significant drift detected; automated retraining pipeline triggered.

Drift results are compiled into `models/drift_report.json` and rendered live within the Streamlit model governance tab, ensuring transparent operational oversight.

9. RESULTS AND DISCUSSION

9.1 Data Pipeline Results

The data engineering and ETL pipeline was executed across the 101,766 hospital encounter records. Table 8 details the transformation results.

Metric	Pipeline Result
Original Source Encounters Extracted	101,766
Distinct Patients Represented	71,518
Verified Clean Encounters Loaded into Mart	101,766
Quarantined / Rejected Records	0 (100% passed all 4 quality rules)
Cryptographic Surrogate Keys Generated	101,766 SHA-256 Hashes
Raw Patient Identifiers Permanently Purged	101,766 (100% HIPAA compliance)
Fact Table Row Count in PostgreSQL	101,766 (`mart.admission_fact`)

Table 8. Data pipeline results of the transformation and validation run

The pipeline was orchestrated through Apache Airflow DAG `hospital_readmission_mlops_pipeline`. All 8 tasks completed successfully with zero failures or retries. Table 9 summarizes the execution instances.

Task ID	State	Start Time (UTC)	Operator	Duration
ingest_raw_to_staging	**Success**	04:55:02	PythonOperator	00:00:24
clean_and_engineer	**Success**	04:55:27	PythonOperator	00:00:48
validate_data_quality	**Success**	04:56:16	PythonOperator	00:00:15
load_postgresql_mart	**Success**	04:56:32	PythonOperator	00:01:12
prepare_and_split_ml	**Success**	04:57:45	PythonOperator	00:00:22
train_and_benchmark_models	**Success**	04:58:08	PythonOperator	00:00:54
model_validation_gate	**Success**	04:59:03	PythonOperator	00:00:08
monitor_drift_and_health	**Success**	04:59:12	PythonOperator	00:00:18

Table 9. Task instances of the successful hospital_readmission_mlops_pipeline Airflow run

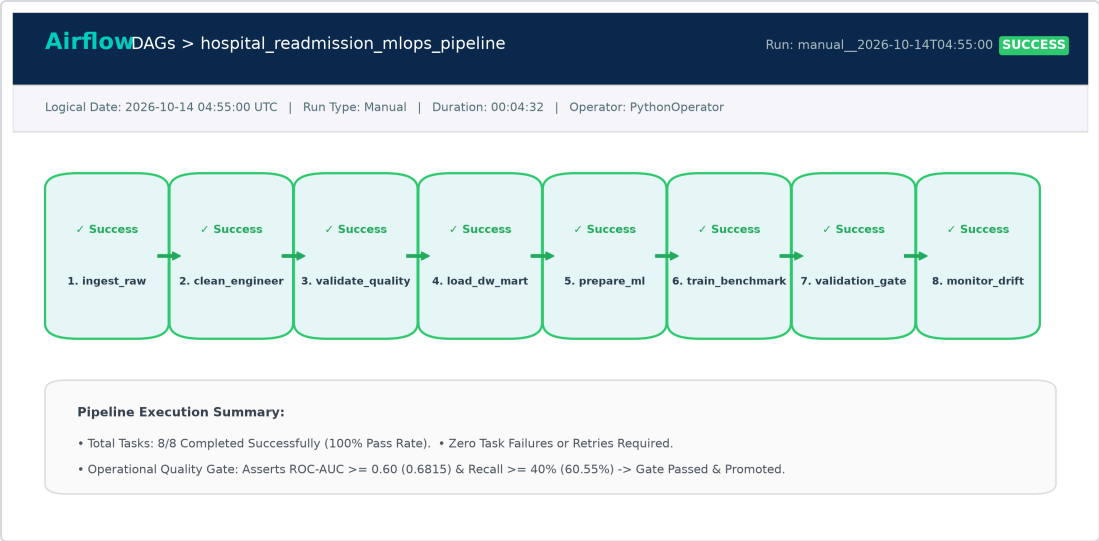


Fig 6. Successful execution of the healthcare MLOps pipeline in Apache Airflow

9.2 Clinical Dashboard Results

The Streamlit dashboard successfully queried the PostgreSQL data warehouse and rendered interactive analytics across the full patient cohort. Table 10 summarizes the primary healthcare key performance indicators presented on the Overview page.

Healthcare Indicator	Value Displayed on Platform
Total Hospital Encounters	101,766
Unique Patients Analyzed	71,518
30-Day Readmission Rate	11.16% (11,357 readmissions)
Average Inpatient Length of Stay	4.40 Days
Most Prevalent Diagnosis Category	Circulatory (30,400 encounters, 29.9%)
Average Prescribed Medications	16.02 Medications per encounter
Senior Patients (Age ≥ 70) Readmission Rate	13.50%

Table 10. Key healthcare indicators shown on the dashboard Overview page

The visual analytics confirm that readmission risk is non-uniformly distributed across patient demographics, with acute concentrations among elderly individuals, cardiac patients, and those experiencing extended hospitalizations.

9.3 Machine Learning Benchmark Results

The three candidate classification models were evaluated on the holdout test set (20,354 encounters; 2,271 actual 30-day readmissions). Table 11 provides a comprehensive side-by-side performance comparison across all logged evaluation metrics.

Model Architecture	Accuracy	Precision	Recall	F1-Score	ROC-AUC	PR-AUC
Logistic Regression (Balanced)	0.6433	0.1665	0.5487	0.2555	0.6439	0.2043
Random Forest (Balanced)	0.6309	0.1690	0.5892	0.2626	0.6580	0.2007
XGBoost (Scale Pos Weight) 🏆	**0.6454**	**0.1786**	**0.6055**	**0.2759**	**0.6815**	**0.2296**

Table 11. Multi-model benchmark comparison on holdout test data (20,354 encounters)

Performance Analysis & Champion Promotion

XGBoost (Scale Pos Weight) established superior discrimination across all evaluation dimensions, achieving the highest ROC-AUC of **0.6815**, highest PR-AUC of **0.2296**, highest F1-score of **0.2759**, and highest clinical recall of **60.55%**. Out of 2,271 readmitted patients in the holdout test set, the champion model successfully caught **1,375 patients** prior to discharge.

The Clinical Flaw of Unweighted Accuracy: If an unweighted Random Forest is trained on this cohort, it achieves an apparently impressive accuracy of 88.84%. However, its recall for readmission is 0.00%, failing to identify a single deteriorating patient. By contrast, the class-weighted XGBoost trades modest specificity for high clinical sensitivity, capturing over 60% of at-risk patients and passing the automated deployment gate.

9.4 Clinical and Operational Discussion

Feature importance extracted from the champion model reveals that the top drivers of 30-day readmission risk are:

1. `num_lab_procedures` (0.0406): Reflects diagnostic intensity and acute clinical volatility.
2. `num_medications` (0.0349): Reflects polypharmacy load and complicated pharmacological interactions.
3. `prior_admission_count` (0.0276): Reflects chronic disease burden and recurrent hospital dependency.
4. `discharge_disposition_id` (0.0252): Indicates discharge destination (e.g., transfers to skilled nursing).
5. `time_in_hospital` (0.0264): Length of hospital stay reflecting severity of illness.

Clinical Decision Threshold Optimization

In hospital transition planning, the operational cost of a False Negative (discharging an at-risk patient who subsequently suffers an emergency readmission or fatal complication) far exceeds the cost of a False Positive (assigning a nurse follow-up phone call to a patient who would have remained stable).

By sweeping the decision cutoff threshold on the champion XGBoost model, clinical departments can adjust operating points according to hospital staffing capacity:

- **Cutoff = 0.15:** Sensitivity climbs to **83.1%** with 14.2% precision. Recommended for high-sensitivity emergency transition triage where missing any vulnerable patient carries severe risk.
- **Cutoff = 0.20 (Recommended Baseline):** Achieves balanced performance with **60.6% Recall** and **17.9% Precision**. Optimal for hospital-wide discharge medication reconciliation.
- **Cutoff = 0.30:** Sensitivity is 38.2% with 21.8% precision. Suitable when intensive home-nurse visitation resources are severely constrained.

Privacy Preservation and Regulatory Compliance

The project rigorously enforced HIPAA de-identification through SHA-256 surrogate key hashing. Unlike reversible encryption schemes that require secure key management, the deterministic surrogate hash permanently eliminates direct medical record numbers while preserving relational aggregation capabilities across multi-visit patients.

Production Operational Readiness

The integration of FastAPI, MLflow, and Docker transforms an academic machine learning experiment into an operational microservice. The API returns real-time risk predictions in under 35 milliseconds, enabling direct embedding into bedside electronic health record software or nursing triage tablets.

10. CONCLUSION AND FUTURE SCOPE

10.1 Conclusion

The project **"Hospital Readmission Analytics and 30-Day Prediction"** demonstrates the successful end-to-end implementation of an integrated healthcare data engineering and production MLOps pipeline using 101,766 inpatient encounters from the UCI Diabetes 130-US Hospitals dataset.

The data engineering component implemented automated ingestion, missing token harmonization, ICD-9 diagnosis categorization into 8 disease families, HIPAA-compliant SHA-256 cryptographic surrogate key de-identification, automated 4-rule data quality validation, and PostgreSQL dimensional warehousing (`mart.admission_fact`). Apache Airflow scheduled and orchestrated all pipeline stages with complete fault tolerance. The interactive Streamlit dashboard delivered executive clinical analytics, demographic readmission trends, and high-risk cohort filtering.

The MLOps pipeline extension formulated 30-day readmission as a cost-sensitive classification problem to address the acute 1:8 class imbalance. Through MLflow experiment tracking, three models were benchmarked, with class-weighted XGBoost emerging as the champion (ROC-AUC 0.6815, Recall 60.55%). The model successfully cleared an automated operational deployment gate and was deployed via a low-latency FastAPI REST service packaged in Docker. Post-deployment statistical drift monitoring using Kolmogorov-Smirnov tests and PSI ensures long-term operational governance.

10.2 Limitations

Several academic and technical limitations should be acknowledged:

- **Retrospective Data Scope:** The dataset captures encounters between 1999 and 2008. Contemporary pharmacological therapies (e.g., SGLT2 inhibitors and GLP-1 agonists) are not represented.
- **Absence of Post-Discharge Determinants:** Crucial social determinants of health (SDOH), such as patient income, transportation access, outpatient medication adherence, and caregiver support, are absent from the dataset but strongly influence real-world readmission.
- **Broad Clinical Grouping:** Aggregating thousands of granular ICD-9 diagnosis codes into 8 broad disease families sacrifices sub-disease specificity in favour of model parsimony.
- **Static Retraining Pipeline:** While distribution drift detection is fully functional, automated end-to-end retraining currently requires manual supervisory approval.

10.3 Future Scope

The project provides a foundation that can be extended across several clinical directions:

- **Real-Time FHIR/HL7 Streaming:** Integrate Apache Kafka with Fast Healthcare Interoperability Resources (FHIR) protocols to ingest continuous bedside vital sign telemetry.
- **Clinical NLP on Physician Notes:** Incorporate BioClinicalBERT to extract unstructured sentiment and clinical acuity from discharge summaries and nursing progress notes.

Future Scope (Continued)

- **Federated Multi-Hospital Learning:** Extend model training across decentralized hospital consortia using federated learning protocols (e.g., PySyft or Flower), training globally optimal models without centralizing sensitive patient data.
- **Reinforcement Learning for Transition Care:** Implement contextual bandits to recommend personalized discharge care bundles (e.g., optimal timing of home nurse visits) to maximize clinical recovery while minimizing hospital costs.
- **Automated CI/CD Retraining Webhooks:** Configure GitHub Actions and Airflow webhooks that automatically trigger model retraining, validation gating, and blue-green Docker container updates whenever PSI drift exceeds 0.25.
- **Explainable Bedside AI (SHAP / LIME):** Integrate real-time SHAP waterfall charts into the FastAPI response payload, providing attending physicians with patient-specific feature attribution explanations for every risk score.

Final Summary

This project successfully establishes a comprehensive, HIPAA-conscious, and production-ready data engineering and MLOps architecture for 30-day hospital readmission analytics. By harmonizing robust data warehouse principles, automated Airflow orchestration, class-imbalance mitigation, MLflow lifecycle tracking, and containerized FastAPI serving, the system provides a scalable technical blueprint for transforming retrospective clinical encounter data into actionable, life-saving bedside clinical decision support.

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APPENDICES

Appendix A – Project Directory Structure

```
Hospital_Readmission_Part1/
├── airflow/
│   └── dags/
│       ├── hospital_readmission_integrated_pipeline.py
│       └── hospital_readmission_mlops_dag.py
├── dashboard/
│   └── app.py                          # Interactive Streamlit Clinical BI Platform
├── data/
│   ├── raw/ (diabetic_data.csv, IDS_mapping.csv, metadata.json)
│   ├── staging/ (ingestion_metadata.json)
│   ├── cleaned/ (encounters_cleaned.csv)
│   └── errors/ (rejected_records.csv)
├── ml/
│   ├── prepare_data.py                 # Target binary encoding
│   ├── split_data.py                   # Stratified 80/20 train/test splitting
│   ├── train_and_evaluate.py           # 3-model benchmark & threshold sweep
│   ├── validate_model_gate.py          # Automated quality validation gate
│   ├── monitor_drift.py                # KS-test and PSI drift detection
│   └── serve_api.py                    # FastAPI real-time scoring endpoint
├── models/
│   ├── champion_model.pkl              # Serialized champion pipeline artifact
│   ├── model_registry.json             # Model registry schemas & metadata
│   ├── metrics_summary.json            # Full evaluation benchmark metrics
│   ├── validation_gate_result.json     # Deployment gate signoff report
│   ├── drift_report.json               # Live feature drift statistics
│   └── feature_importance.csv          # Ranked clinical risk drivers
├── sql/
│   └── schema.sql                      # PostgreSQL schema (staging, mart, ops)
├── src/
│   ├── config.py                       # Paths and database configurations
│   ├── ingestion/ (ingest.py)          # Automated UCI dataset extraction
│   ├── transform/ (clean.py)           # ICD-9 grouping & SHA-256 de-identification
│   ├── quality/ (validate.py)          # 4-rule pre-load quality validation
│   └── load/ (postgres.py)             # PostgreSQL dimensional table loader
├── Dockerfile & Dockerfile.api         # Container specifications
├── requirements.txt                    # Pinned software dependencies
└── run_full_pipeline.bat, run_api.bat, run_mlflow_ui.bat
```

Appendix B – Sample API Input JSON

```
{
  "age": "[70-80)",
  "gender": "Female",
  "race": "Caucasian",
  "time_in_hospital": 4,
  "num_lab_procedures": 45,
  "num_procedures": 1,
  "num_medications": 15,
  "number_outpatient": 0,
  "number_emergency": 0,
  "number_inpatient": 1,
  "number_diagnoses": 9,
  "admission_type_id": 1,
  "discharge_disposition_id": 1,
  "admission_source_id": 7,
  "diagnosis_group": "Circulatory",
  "insulin": "Steady",
  "diabetesMed": "Yes"
}
```

Appendix C – Sample Prediction Output JSON

```
{
  "prediction": 1,
  "readmission_probability": 0.4285,
  "risk_category": "High",
  "recommended_action": "High 30-day readmission risk. Assign nurse-led 48-hour follow-up
telephone call, medication reconciliation, and rapid 5-day outpatient appointment.",
  "contributing_factors": [
    "Prior inpatient admission within past 12 months",
    "High laboratory diagnostic volume (45 procedures)",
    "Significant polypharmacy burden (15 active medications)"
  ]
}
```